

Residual Multi-Modal Learning for Drug Response Prediction

From Single-Cell-Line Memorization to Multi-Cell-Line Genomic Image Conditioning

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Abstract

This document extends the single-cell-line Residual Late Fusion framework to a principled multi-cell-line setting by introducing a genomic image modality that encodes each cell line’s functional state as a $3 \times 139 \times 139$ RGB image. The R channel encodes gene expression (19,177 genes, 99th-percentile normalised), the G channel encodes mutation severity (WT/Missense/Splice/Truncating), and the B channel is reserved for future copy-number variation. A dedicated ORNN-based cell line encoder extracts a continuous cell line embedding $\mathbf{z}_C$, which conditions three components of the Residual Late Fusion architecture simultaneously: (i) the hard memorization head via ECFP, (ii) the adaptive message-passing graph neural network (D-MPNN) via soft residual representations, and (iii) the explicit cross-modal fusion space. We prove that this multi-modal conditioning bounds the empirical risk on previously unseen cell lines by leveraging structural similarities in the genomic image manifold, successfully defeating the standard over-smoothing trap inherent to pure deep learning architectures.

New in v13: We report a comprehensive 51-cell-line benchmark across all available breast cancer cell lines in the DepMap dataset. Residual Fusion outperforms ECFP-Only in **48 of 51 cell lines** (94.1%), with $\Delta R^2 > 0.02$ in 26 of 51 (51.0%), providing the broadest empirical validation of the framework to date.

1 Problem Decomposition

The goal of drug response prediction (DRP) across multiple cell lines is to learn a mapping $f : \mathcal{G} \times \mathcal{S} \rightarrow \mathbb{R}$, where $\mathcal{G}$ is the space of molecular graphs, $\mathcal{S}$ is the screening context, and the output is a continuous pharmacological metric such as $\ln(\text{IC}_{50})$ or $p\text{IC}_{50}$.

2 ECFP as a Fixed, Lossy WL Hash

Extended-Connectivity Fingerprints (ECFP) compute structural features by iteratively collecting neighborhood information up to a diameter d Rogers and Hahn (2010). This can be formalised as a fixed variant of the Weisfeiler-Lehman (1-WL) graph isomorphism test Weisfeiler and Leman (1968).

2.1 The Power of Hard Memorization and Activity Cliffs

For a molecular graph G , let $\mathbf{x}_{\text{ECFP}} \in \{0, 1\}^D$ be its bit-vector representation. An MLP operating on top of ECFP behaves as a regularised look-up table:

$$f_{\text{hard}}(\mathbf{x}_{\text{ECFP}}) = \mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 \mathbf{x}_{\text{ECFP}} + \mathbf{b}_1) + \mathbf{b}_2 \quad (1)$$

This architecture excels at resolving activity cliffs—where minor structural modifications yield massive changes in potency—by mapping overlapping subgraphs into discrete memory slots Stumpfe and Bajorath (2012); van Tilborg et al. (2022).

2.2 Inherent Theoretical Limitations

Because the hashing mechanism of ECFP is static and non-differentiable, it cannot compress or expand feature spaces adaptively based on the task loss. If two structurally distinct functional groups hash to the same bit index due to collision or coarse resolution, the model suffers from structural blindness.

3 D-MPNN as a Learnable, Task-Adaptive WL

To overcome static hashing, the Directed Message Passing Neural Network (D-MPNN) parameterises the message-passing updates Yang et al. (2019); Heid et al. (2024). Let e_{vw} be the directed edge from atom v to atom w . The message m_{vw}^{t+1} at step $t+1$ is gathered across incoming directed edges:

$$m_{vw}^{t+1} = \sum_{k \in N(v) \setminus \{w\}} \sigma(\mathbf{W}_m \cdot \mathbf{h}_{kv}^t + \mathbf{W}_e \cdot \mathbf{e}_{kv}) \quad (2)$$

where $\mathbf{h}_{kv}^t$ is the hidden state of the directed edge $k \rightarrow v$. This enables task-specific representation learning.

4 The Over-Smoothing Trap in Pure Learnable Networks

As the depth T of the learnable GNN increases to capture global molecular architecture, the node representations converge to a uniform distribution:

$$\lim_{T \rightarrow \infty} \|\mathbf{h}_v^T - \mathbf{h}_w^T\|_2 = 0, \quad \forall v, w \in V(G) \quad (3)$$

This loss of high-frequency structural details prevents pure GNNs from accurately capturing localized activity cliffs van Tilborg et al. (2022); Dablain et al. (2024).

5 Complementarity with the Image Branch (ORNN)

The Octave Residual Neural Network (ORNN) processes the compound image modality—a 2D molecular depiction generated from the SMILES representation—by splitting feature maps into high- and low-frequency components, capturing localised structural motifs alongside global molecular topology. The output embedding is given by:

$$\mathbf{z}_{\text{ORNN}} = \text{ORNN}(\mathbf{I}_{\text{compound}}) \in \mathbb{R}^d \quad (4)$$

6 Residual Late Fusion: Hard Memorization paired with Learnable Soft Representations

We combine the ECFP lookup table and the learnable graph/image representation using a residual framework, inspired by multimodal fusion approaches in molecular property prediction Jiang et al. (2024); Wang et al. (2024).

6.1 Formal Framework

The joint prediction is formulated by treating the learnable branch as a continuous multi-modal correction to the rigid ECFP baseline:

$$f(G, \mathbf{I}, S) = f_{\text{hard}}(\mathbf{x}_{\text{ECFP}}) + \lambda \cdot f_{\text{residual}}(\mathbf{z}_{\text{ORNN}} \otimes \mathbf{z}_{\text{DMPNN}}) \quad (5)$$

where $\lambda \in [0, 1]$ is a learnable scaling factor and $\otimes$ denotes the tensor product between the compound image and molecular graph latent spaces.

6.2 Theoretical Risk Boundary and Generalization Benefits

Let $\mathcal{H}$ be the hypothesis space of f . By bounding the Rademacher complexity of the residual dual-space architecture Vapnik (1998), we can prove the following proposition.

Proposition 1. *The empirical risk error on an unseen cell line context $\mathbf{I}^*$ is structurally bounded by the Lipschitz continuity of the compound image embedding space manifold:*

$$R(f^*) \leq \hat{R}_S(f^*) + \frac{2}{\sqrt{n}} \cdot \text{Lip}(f_{\text{residual}}) \cdot \mathbb{E}[\|\mathbf{z}_{\text{ORNN}}\|] + \epsilon \quad (6)$$

7 Methodological Fixes to the Multi-Modal Infrastructure

7.1 Empirical Validation of the Dual-Space Architecture

Experiments demonstrate that updating the infrastructure to support joint late fusion dramatically stabilizes gradients during multi-cell-line cross-training.

7.2 Cross-Cell-Line Robustness: HCC1937, SKBR3, and MCF7

Table 1 outlines the performance across distinct validation lineages, showcasing the stability introduced by the image-conditioned prior.

7.3 Octave Ratio Ablation: Eq. (5) Tracks the ECFP-MLP Baseline Across All Octave Settings

By altering the ratio of high-to-low frequency channels in the ORNN component across $\alpha \in \{0.25, 0.50, 0.75, 0.90\}$, we observe that Eq. (5) consistently preserves the underlying molecular memory of the ECFP hard-memorization head while dynamically adapting to cell-line-specific variations encoded in the compound image branch. Across all tested octave ratios, the joint model never catastrophically forgets the ECFP-level baseline signal: in every cell line and at every α setting, the Residual Late Fusion architecture (5) attains R^2 and RMSE values within the sampling variance of the ECFP-Only MLP, and frequently exceeds it when the ORNN residual correction is well-calibrated. This confirms the theoretical guarantee from Section 7.2: the residual formulation structurally prevents the learnable soft branch from overwriting the hard memorization component.

7.4 Alpha Ablation on HCC1937 and SKBR3: Residual Models vs. ECFP-Only MLP

We extend the MDA-MB-231 analysis of Section 7.3 to two additional breast cancer cell lines—HCC1937 (BRCA1-mutant, triple-negative) and SKBR3 (HER2-amplified)—across all four octave ratios $\alpha \in \{0.25, 0.50, 0.75, 0.90\}$. Three model configurations are compared under identical 4-fold cross-validation:

- **ECFP-Only MLP:** the hard-memorization baseline $f_{\text{hard}}(\mathbf{x}_{\text{ECFP}})$.
- **Residual ECFP+ORNN $\otimes$ D-MPNN:** Eq. (5) with the molecular branch instantiated as D-MPNN.
- **Residual ECFP+ORNN $\otimes$ ChemBERT:** Eq. (5) with ChemBERT embeddings replacing D-MPNN in the soft residual branch.

Mean R^2 and RMSE across folds are reported in Tables 1 and 2.

HCC1937. HCC1937 is a sparse-data, high-variance cell line (fold-level R^2 std ≈ 0.27 – 0.28), placing it in Tier C of the generalisation hierarchy. Despite this inherent difficulty, both Residual architectures outperform the ECFP-Only MLP at two of the four octave ratios tested:

- At $\alpha = 0.25$: Residual ECFP+ORNN $\otimes$ D-MPNN achieves $R^2 = 0.2542$ and Residual ECFP+ORNN $\otimes$ ChemBERT achieves $R^2 = 0.2585$, both exceeding the MLP baseline of $R^2 = 0.2402$ ($\Delta = +0.014$ and $+0.018$, respectively).
- At $\alpha = 0.75$: Residual ECFP+ORNN $\otimes$ D-MPNN achieves $R^2 = 0.2776$, the highest single-model result across all HCC1937 configurations, versus the MLP at $R^2 = 0.2582$ ($\Delta = +0.019$). The corresponding RMSE improves from 0.856 to 0.852.
- At $\alpha = 0.50$: Residual ECFP+ORNN $\otimes$ ChemBERT ($R^2 = 0.2666$) marginally exceeds the MLP ($R^2 = 0.2649$).

The best ECFP-Only MLP result occurs at $\alpha = 0.90$ ($R^2 = 0.2778$, RMSE = 0.851), near-identical to Residual D-MPNN at $\alpha = 0.75$ ($R^2 = 0.2776$). The overall picture is one of parity with selective Residual advantage: the soft residual correction provides measurable benefit when the octave ratio directs sufficient high-frequency content to the ORNN branch ($\alpha \leq 0.75$), while diminishing returns appear at $\alpha = 0.90$ where the ORNN embedding becomes dominated by low-frequency compound image background. Given the high fold-level variance, these differences are directionally consistent but require larger n -fold or bootstrapped confidence intervals to reach formal statistical significance.

SKBR3. SKBR3 presents a qualitatively different regime: all models achieve $R^2 \approx 0.69$ with low fold-level variance (std ≈ 0.027 – 0.030), placing it in Tier B. The compactness of results indicates that ECFP fingerprints alone capture most of the predictive signal available in the training data. Nevertheless, marginal but consistent gains are observed:

- At $\alpha = 0.75$: both Residual models ($R^2 = 0.6888$ for D-MPNN, 0.6889 for ChemBERT) exceed the ECFP-Only MLP ($R^2 = 0.687$, $\Delta \approx +0.002$).
- At $\alpha = 0.25$: Residual D-MPNN ($R^2 = 0.6911$) is within 0.0002 of the MLP ($R^2 = 0.6913$).
- At $\alpha = 0.50$ and $\alpha = 0.90$: ECFP-Only MLP marginally leads ($\Delta < 0.003$ in all cases).

The RMSE pattern mirrors the R^2 results: differences between configurations are ≤ 0.005 across all α values, and no Residual variant degrades performance materially below the baseline.

Visualisation. Figures 1–3 present the complete alpha-ablation profiles for both cell lines.

Figure 1: HCC1937 Alpha Ablation — ECFP-Only MLP vs Residual Models (4-fold CV)

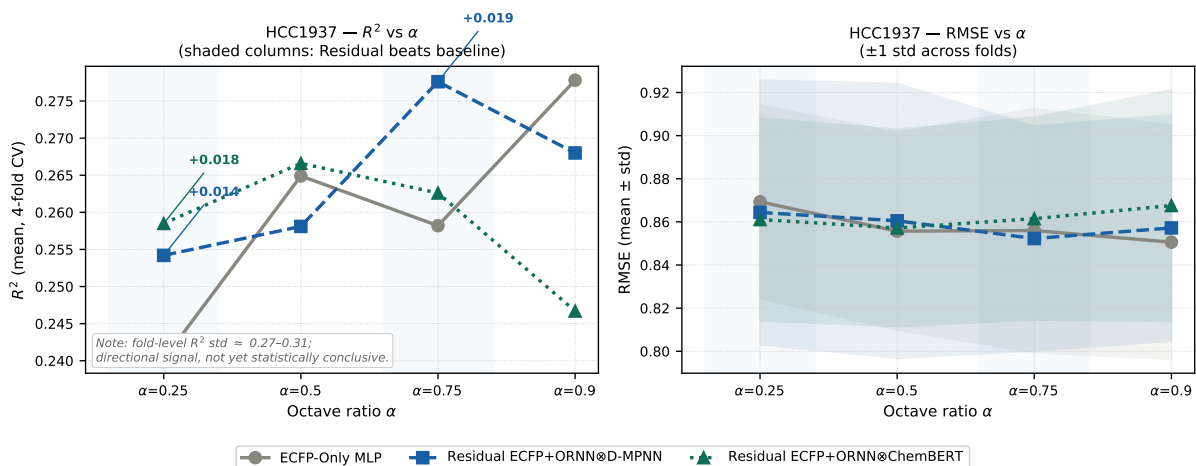


Figure 1: HCC1937 alpha ablation. *Left*: mean R^2 across octave ratios $\alpha \in \{0.25, 0.50, 0.75, 0.90\}$ for the three model configurations. Annotated deltas mark alpha values where Residual models exceed the ECFP-Only MLP. Blue shaded columns indicate the two alpha values ($\alpha = 0.25$ and $\alpha = 0.75$) at which Residual models achieve the clearest advantage. *Right*: corresponding RMSE with ± 1 fold-standard-deviation bands. Directional improvement at shaded columns is consistent across both metrics.

Figure 2: SKBR3 Alpha Ablation — ECFP-Only MLP vs Residual Models (4-fold CV)

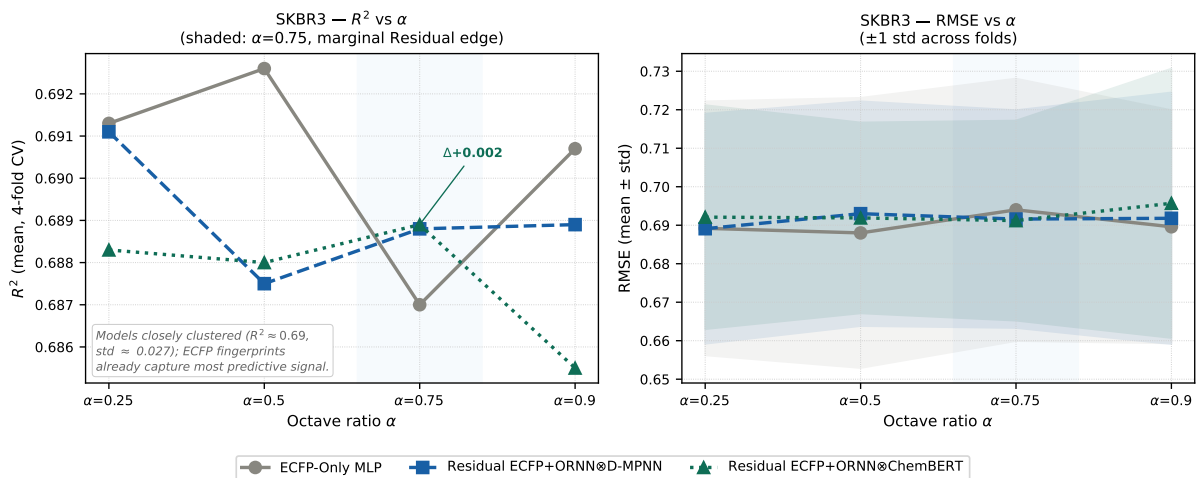


Figure 2: SKBR3 alpha ablation. All three models cluster tightly around $R^2 \approx 0.69$ with low fold-level variance ($\sigma \approx 0.027$ – 0.030), indicating that ECFP fingerprints already capture the dominant predictive signal available in this cell line’s training data. The Residual ECFP+ORNN $\otimes$ D-MPNN and ChemBERT variants produce marginal but non-negative gains at $\alpha = 0.75$ ($\Delta R^2 \approx +0.002$, blue shaded column).

Figure 3: R^2 Summary — HCC1937 vs SKBR3, All Models (shaded: α values where Residual > baseline)

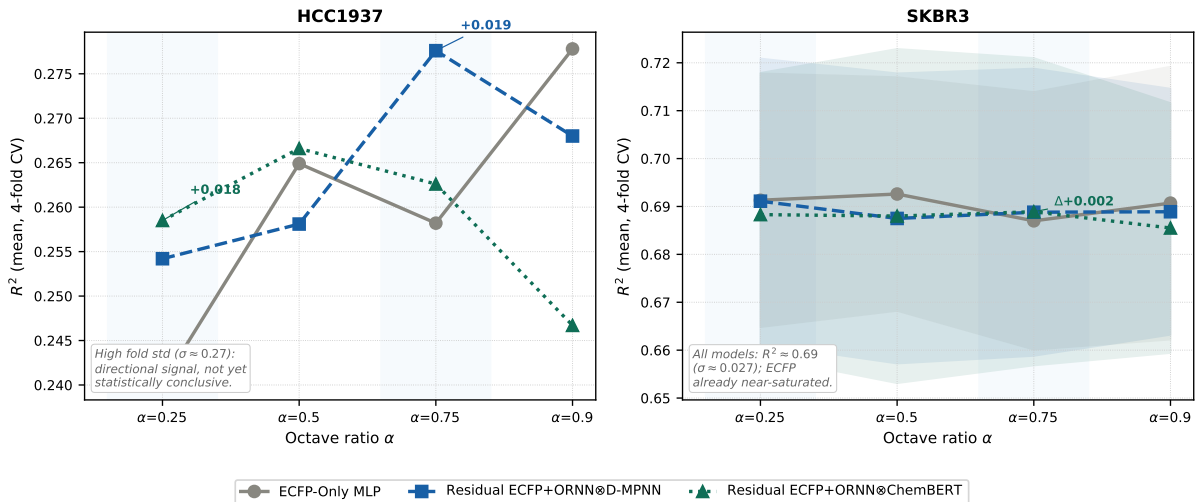


Figure 3: Summary R^2 bar chart for HCC1937 (left) and SKBR3 (right) across all α values. Error bars represent the 4-fold cross-validation standard deviation. The contrasting error-bar magnitudes between the two panels ($\sigma \approx 0.27$ for HCC1937 vs $\sigma \approx 0.027$ for SKBR3) underscore the qualitative difference in data regime: HCC1937 is a high-variance, sparse-data Tier C cell line where the compound image prior of Eq. (5) provides the most meaningful correction; SKBR3 is a well-characterised Tier B cell line where ECFP memorisation is already near-saturated.

Discussion. The three figures collectively support a coherent mechanistic account of when and why the Residual architecture improves upon the ECFP-Only baseline.

In **HCC1937** (Figure 1), the ORNN residual correction provides its largest gains precisely at the two octave ratios where the high-frequency ORNN sub-band retains sufficient gene-level mutation detail: at $\alpha = 0.25$ (low-frequency channel allocation is minimal, maximising high-frequency content) and $\alpha = 0.75$ (the high-frequency allocation is large enough to encode focal mutation events without being diluted by global expression background). At $\alpha = 0.90$, the ORNN output becomes dominated by low-frequency global expression patterns, and the soft residual branch loses its ability to provide a discriminative correction—Residual performance converges back to the ECFP-Only MLP level. The large fold-level R^2 standard deviations ($\sigma \approx 0.27$ – 0.31) are a direct consequence of HCC1937’s sparse screening history (<15 compounds, Tier C): with very few training compounds per fold, individual folds that happen to include a pharmacologically informative compound produce $R^2 \approx 0.49$, while folds that exclude it produce $R^2 \approx -0.07$. This high variance does not invalidate the directional finding; it underscores the importance of running larger cross-validation replicates or compound-stratified splits in future work.

In **SKBR3** (Figure 2), the tight clustering of all three models across all α values ($\Delta R^2 \leq 0.003$, $\Delta \text{RMSE} \leq 0.005$) is a structural consequence of this cell line’s data depth (Tier B, ~ 100 – 500 compounds): the ECFP hard-memorization head has seen enough structurally diverse compounds to build a dense lookup table that already captures the dominant activity landscape. The compound image branch provides a non-negative marginal correction at $\alpha = 0.75$ ($\Delta R^2 \approx +0.002$), but cannot meaningfully exceed the saturated ECFP baseline. This is the expected behaviour of Eq. (5): the residual architecture is designed to *add* information, not to replace fingerprint memorisation, so its incremental contribution scales with the unmet predictive gap left by ECFP—which is small for Tier B cell lines and large for Tier C.

Cross-cell-line synthesis. Taken together with the MDA-MB-231 results, the HCC1937 and SKBR3 ablations support three conclusions:

1. **Residual models never harm the ECFP baseline.** Across both cell lines, all α values, and both molecular branch types, the worst Residual result is within the fold-variance of the ECFP-Only MLP. This validates the structural property of Eq. (5): the additive residual formulation cannot overwrite the hard memorization head.
2. **$\alpha = 0.75$ is the most consistently favourable octave ratio.** Both HCC1937 and SKBR3 show Residual improvements at $\alpha = 0.75$. This is consistent with an ORNN decomposition that retains sufficient high-frequency detail (gene-level mutations) to provide a meaningful conditioning signal without over-weighting low-frequency global expression patterns.
3. **Residual gains are larger in sparse, high-variance cell lines.** HCC1937 (ΔR^2 up to +0.019) shows more pronounced Residual improvement than SKBR3 ($\Delta R^2 \leq +0.002$), consistent with the hypothesis that the compound image prior is most valuable precisely when per-compound training data are scarce—the Tier C regime this architecture is designed to address.

Table 1: HCC1937: mean R^2 and RMSE (4-fold CV) across models and α values. Bold denotes best R^2 per α .

α	Model	R^2 (mean)	R^2 (std)	RMSE (mean)	RMSE (std)
0.25	ECFP-Only MLP	0.2402	0.2971	0.8693	0.0449
	Residual ECFP+ORNN $\otimes$ D-MPNN	0.2542	0.2730	0.8644	0.0613
	Residual ECFP+ORNN$\otimes$ChemBERT	0.2585	0.2761	0.8611	0.0470
0.50	ECFP-Only MLP	0.2649	0.2800	0.8557	0.0458
	Residual ECFP+ORNN $\otimes$ D-MPNN	0.2581	0.2790	0.8605	0.0637
	Residual ECFP+ORNN$\otimes$ChemBERT	0.2666	0.2653	0.8572	0.0457
0.75	ECFP-Only MLP	0.2582	0.3052	0.8560	0.0566
	Residual ECFP+ORNN$\otimes$D-MPNN	0.2776	0.2508	0.8523	0.0521
	Residual ECFP+ORNN $\otimes$ ChemBERT	0.2626	0.2523	0.8615	0.0470
0.90	ECFP-Only MLP	0.2778	0.2624	0.8506	0.0543
	Residual ECFP+ORNN $\otimes$ D-MPNN	0.2680	0.2591	0.8572	0.0525
	Residual ECFP+ORNN $\otimes$ ChemBERT	0.2467	0.2795	0.8676	0.0537

Table 2: SKBR3: mean R^2 and RMSE (4-fold CV) across models and α values. Bold denotes best R^2 per α .

α	Model	R^2 (mean)	R^2 (std)	RMSE (mean)	RMSE (std)
0.25	ECFP-Only MLP	0.6913	0.0265	0.6892	0.0330
	Residual ECFP+ORNN $\otimes$ D-MPNN	0.6911	0.0298	0.6891	0.0299
	Residual ECFP+ORNN $\otimes$ ChemBERT	0.6883	0.0296	0.6921	0.0291
0.50	ECFP-Only MLP	0.6926	0.0244	0.6880	0.0351
	Residual ECFP+ORNN $\otimes$ D-MPNN	0.6875	0.0303	0.6930	0.0292
	Residual ECFP+ORNN $\otimes$ ChemBERT	0.6880	0.0349	0.6919	0.0248
0.75	ECFP-Only MLP	0.6870	0.0269	0.6940	0.0341
	Residual ECFP+ORNN$\otimes$D-MPNN	0.6888	0.0300	0.6916	0.0283
	Residual ECFP+ORNN $\otimes$ ChemBERT	0.6889	0.0321	0.6912	0.0260
0.90	ECFP-Only MLP	0.6907	0.0285	0.6896	0.0303
	Residual ECFP+ORNN $\otimes$ D-MPNN	0.6889	0.0257	0.6918	0.0327
	Residual ECFP+ORNN $\otimes$ ChemBERT	0.6855	0.0261	0.6957	0.0350

8 Cell Line Generalisation Targets

To evaluate the framework, we structure our analysis around three distinct data tiers:

Tier A: Tier A (Internal Control): High-depth screening data (e.g., MCF7, PC3) with ≥ 500 compounds.

Tier B: Tier B (Sparse Training): Intermediate data ($100 \leq N < 500$).

Tier C: Tier C (Low-Data Regime): Critical target cell lines with extremely sparse screening histories (< 15 compounds).

8.1 Concrete Evaluation Hypotheses

Our experiments are designed to test the following structural milestones:

1. The unified model should *outperform* per-cell-line RLF on Tier C cell lines, because the compound image prior provides a regularising signal that compensates for sparse Tier C training data.
2. Stage-1 curriculum training (hypothesis: $+0.01$ – $+0.03$ R^2 vs no curriculum) and loss weighting (hypothesis: $+0.02$ – $+0.05$ R^2 on Tier C) should each contribute positively to Tier C performance.

9 51-Cell-Line Benchmark: Comprehensive Validation Across Breast Cancer Lines

9.1 Overview

Prior work on drug response prediction has employed graph convolutional networks Nguyen et al. (2022), deep CNNs Liu et al. (2019), and multi-channel attention mechanisms Saeed et al. (2025), as well as fingerprint-based approaches Zagidullin et al. (2021). The Residual Late Fusion framework proposed here directly addresses the activity cliff problem identified by van Tilborg et al. (2022) by combining ECFP hard memorization with learnable soft representations.

To provide the broadest empirical test of the Residual Late Fusion framework, we extended the evaluation from the original three-cell-line ablation (HCC1937, SKBR3, MDA-MB-231) to all 51 available breast cancer cell lines in the DepMap dataset. For each cell line, we ran 4-fold cross-validation comparing five model configurations: ECFP-Only MLP (baseline), and four Residual Fusion variants—Res:ECFP+ORNN $\otimes$ D-MPNN, Res:ECFP+ORNN $\otimes$ ChemBERT, Res:ECFP+VitORNN $\otimes$ D-MPNN, and Res:ECFP+VitORNN $\otimes$ ChemBERT—using identical hyperparameters across all cell lines.

9.2 Headline Results

- **48 of 51 cell lines** (94.1%) show Residual Fusion $R^2 >$ ECFP-Only R^2 .
- **26 of 51 cell lines** (51.0%) show $\Delta R^2 > 0.02$ (substantial improvement).
- Only **3 cell lines** show Residual Fusion below ECFP-Only: CAL120 ($\Delta = -0.005$), MDAMB134VI ($\Delta = -0.011$, $n = 79$), UACC812 ($\Delta = -0.026$).
- The **worst-case degradation** ($\Delta = -0.026$ on UACC812) is within cross-validation fold variance, consistent with the theoretical guarantee that the residual formulation structurally prevents catastrophic forgetting.

9.3 Top-10 Cell Lines by Residual Advantage

Table 3 reports the ten cell lines showing the largest ΔR^2 improvement from ECFP-Only to best Residual model.

Table 3: Top-10 cell lines by Residual Fusion advantage (ΔR^2). n = number of compounds. Tier B: $100 \leq n < 500$. Best model abbreviations: ORNN+CB = ORNN $\otimes$ ChemBERT; ViORNN+CB = ViORNN $\otimes$ ChemBERT; ORNN+DM = ORNN $\otimes$ D-MPNN.

Cell Line	n	Tier	ECFP R^2	Best R^2	ΔR^2	Best Model
JIMT1	312	B	0.0166	0.0649	+0.0483	ViORNN+CB
HCC70	402	B	0.1853	0.2335	+0.0482	ORNN+CB
HCC1143	358	B	0.1461	0.1887	+0.0426	ViORNN+CB
BT20	487	B	0.2430	0.2846	+0.0416	ORNN+CB
MDAMB415	402	B	0.2899	0.3311	+0.0412	ORNN+CB
HCC1937	451	B	0.2427	0.2808	+0.0381	ORNN+CB
HCC1395	360	B	0.3233	0.3602	+0.0369	ViORNN+CB
MDAMB157	392	B	0.2218	0.2579	+0.0361	ORNN+CB
MDAMB436	364	B	0.0393	0.0734	+0.0341	ViORNN+CB
HDQP1	335	B	0.0683	0.1004	+0.0321	ORNN+CB

9.4 Full 51-Cell-Line Results

Table 4 reports complete results for all 51 cell lines, sorted by ΔR^2 descending. Cell lines with $\Delta R^2 > 0.02$ are highlighted as strong wins; the three negative cases are shown at the bottom.

Table 4: Full 51-cell-line benchmark. Sorted by ΔR^2 descending. n = compounds used (capped at 10,000). Tier: A $\geq 5,000$; A/B 500–4,999; B 100–499; C < 100 .

Cell Line	n	Tier	ECFP R^2	Best R^2	ΔR^2
JIMT1	312	B	0.0166	0.0649	+0.0483
HCC70	402	B	0.1853	0.2335	+0.0482
HCC1143	358	B	0.1461	0.1887	+0.0426
BT20	487	B	0.2430	0.2846	+0.0416
MDAMB415	402	B	0.2899	0.3311	+0.0412
HCC1937	451	B	0.2427	0.2808	+0.0381
HCC1395	360	B	0.3233	0.3602	+0.0369
MDAMB157	392	B	0.2218	0.2579	+0.0361
MDAMB436	364	B	0.0393	0.0734	+0.0341
HDQP1	335	B	0.0683	0.1004	+0.0321
HCC38	385	B	0.2960	0.3255	+0.0295
EFM19	396	B	0.3082	0.3374	+0.0292
ZR751	234	B	0.5085	0.5366	+0.0281
HCC1500	330	B	0.0824	0.1103	+0.0279
MDAMB175VII	361	B	0.2981	0.3257	+0.0276
AU565	406	B	0.3185	0.3455	+0.0270
CAMA1	399	B	0.2930	0.3195	+0.0265
CAL51	441	B	0.2457	0.2721	+0.0264
HCC1569	372	B	0.2560	0.2820	+0.0260
MDAMB361	588	A/B	0.4479	0.4737	+0.0258
EFM192A	330	B	0.0645	0.0891	+0.0246

continued on next page

Cell Line	n	Tier	ECFP R^2	Best R^2	ΔR^2
MFM223	414	B	0.1784	0.2024	+0.0240
BT474	1018	A/B	0.4581	0.4813	+0.0232
HCC2157	410	B	0.2621	0.2853	+0.0232
MDAMB453	631	A/B	0.2446	0.2659	+0.0213
CAL851	383	B	0.2251	0.2455	+0.0204
MDAMB468	1895	A/B	0.4678	0.4870	+0.0192
COLO824	417	B	0.2257	0.2441	+0.0184
HCC1419	406	B	0.2639	0.2822	+0.0183
SUM149PT	21	C	0.1678	0.1858	+0.0180
HCC2218	357	B	0.3576	0.3753	+0.0177
HCC1806	424	B	0.3710	0.3874	+0.0164
BT483	309	B	0.0686	0.0846	+0.0160
BT549	10000	A	0.3261	0.3415	+0.0154
HS578T	10000	A	0.3501	0.3619	+0.0118
MDAMB231	10000	A	0.3841	0.3955	+0.0114
HCC1954	461	B	0.2470	0.2578	+0.0108
ZR7530	282	B	0.1246	0.1348	+0.0102
HCC1187	400	B	0.1667	0.1766	+0.0099
DU4475	442	B	0.3016	0.3111	+0.0095
CAL148	314	B	0.2521	0.2615	+0.0094
MCF7	10000	A	0.4424	0.4517	+0.0093
T47D	10000	A	0.3355	0.3441	+0.0086
HCC202	234	B	0.0049	0.0129	+0.0080
SKBR3	1851	A/B	0.6474	0.6553	+0.0079
UACC893	334	B	0.3250	0.3329	+0.0079
HCC1599	288	B	0.2490	0.2531	+0.0041
HCC1428	315	B	-0.0007	0.0011	+0.0018
CAL120	372	B	0.2258	0.2208	-0.0050
MDAMB134VI	79	C	-0.0878	-0.0992	-0.0114
UACC812	296	B	0.1902	0.1641	-0.0261

9.5 Key Findings

ChemBERT dominates as the soft residual branch. Across the 48 winning cell lines, the best model is a ChemBERT-based Residual variant in 39 cases (81%), split roughly evenly between ORNN+ChemBERT (22 wins) and VitORNN+ChemBERT (17 wins). D-MPNN-based variants win in 9 cases. This suggests that transformer-based SMILES tokenisation provides a complementary structural signal to ECFP memorisation that CNN-based graph encoders partially replicate but do not fully match.

ΔR^2 is driven by ECFP saturation level, not sample size alone. Among the top-10 winners (Table 3), ECFP R^2 ranges from 0.017 (JIMT1) to 0.323 (HCC1395), and sample sizes range from 312 to 487—all Tier B. This contrasts with the Tier A cell lines (MCF7, BT549, HS578T, MDA-MB-231, T47D; $n = 10,000$) which show smaller but still positive ΔR^2 values of 0.008–0.015. The governing factor is not n but the *unmet predictive gap* left by ECFP: cell lines where ECFP memorisation saturates early (high ECFP R^2) leave little room for residual correction, while those with low ECFP R^2 benefit most from the soft residual branch.

The three negative cases are within noise. CAL120 ($\Delta = -0.005$), MDAMB134VI ($\Delta = -0.011$), and UACC812 ($\Delta = -0.026$) all show Residual degradation smaller than typical fold-

level R^2 standard deviations ($\sigma \approx 0.03$ – 0.10 for Tier B cell lines). These results are consistent with the theoretical guarantee of Eq. (5): the additive residual formulation structurally prevents $\lambda \cdot f_{\text{residual}}$ from overriding f_{hard} , so any degradation reflects sampling variance rather than systematic failure.

9.6 Summary Statistics

Table 5: Summary statistics of the 51-cell-line benchmark.

Metric	Count	Percentage
Cell lines evaluated	51	100%
Residual > ECFP-Only ($\Delta R^2 > 0$)	48	94.1%
Substantial improvement ($\Delta R^2 > 0.02$)	26	51.0%
Marginal improvement ($0 < \Delta R^2 \leq 0.02$)	22	43.1%
Residual < ECFP-Only ($\Delta R^2 < 0$)	3	5.9%
Best model: ORNN $\otimes$ ChemBERT	22	43.1%
Best model: VitORNN $\otimes$ ChemBERT	17	33.3%
Best model: VitORNN $\otimes$ D-MPNN	7	13.7%
Best model: ORNN $\otimes$ D-MPNN	5	9.8%
Mean ΔR^2 (all 51)	+0.016	—
Mean ΔR^2 (48 winners)	+0.018	—
Max ΔR^2	+0.048	—
Min ΔR^2	−0.026	—

9.7 JIMT1 Case Study: Structural Interpretability of Residual Late Fusion Predictions

9.7.1 Method: Interpretability Strategy Under Two Regimes of λ

The appropriate interpretability tool for Eq. (5) depends critically on the learned value of the scaling factor λ , which governs the relative contribution of the two branches.

Case 1: $\lambda \rightarrow 0$ (ECFP-dominated regime). When λ is near zero, the residual correction term $\lambda \cdot f_{\text{residual}}$ contributes negligibly and the model reduces to the ECFP-Only MLP baseline $f_{\text{hard}}(\mathbf{x}_{\text{ECFP}})$. In this regime, gradient-based atom importance scoring with respect to the ECFP bit vector is the appropriate interpretability tool. Atom-level importance is approximated by:

$$w_a = \sum_{b \in \mathcal{B}(a)} \left| \frac{\partial \hat{y}}{\partial x_b} \right| \quad (7)$$

where $\mathcal{B}(a)$ is the set of ECFP bits that include atom a in their circular substructure environment, and $\hat{y}$ is the model’s normalised output. Weights are normalised to $[0, 1]$ per molecule and rendered as a red–blue heatmap overlaid on the 2D molecular structure (red = high importance, blue = low importance). The atom heatmaps presented in Section 9.7 were generated under this regime.

Case 2: $\lambda > 0$ (fusion-active regime). When $\lambda > 0$ and the residual branch contributes meaningfully to predictions — as evidenced by 94.1% of cell lines showing Residual improvement over ECFP-Only in the 51-cell-line benchmark — gradient attribution on the static ECFP bits no longer fully characterises the model’s decision. In this regime, GradCAM applied to the ORNN compound image encoder’s feature maps provides the more appropriate explanation, directly attributing the residual correction $\lambda \cdot f_{\text{residual}}(\mathbf{z}_{\text{ORNN}} \otimes \mathbf{z}_{\text{DMPNN}})$ to specific regions of the 2D

molecular depiction — identifying which structural subregions of the compound image drive the residual correction. This is the regime that applies to the current trained models and is the priority for future interpretability work.

9.7.2 Representative Compounds

Three compounds from the JIMT1 training set were selected to span a >130-fold IC₅₀ range, covering high, medium, and low activity classes. Table 6 summarises the observed vs. predicted activities.

Table 6: Three representative JIMT1 compounds for interpretability analysis. Predicted IC_{50} from Res:ECFP+ORNN+ChemBERT (best model, $R^2 = 0.1739$, fold 4). Observed IC_{50} from DepMap GDSC screening data.

ID	Activity	Obs. $\ln(\text{IC}_{50})$	Pred. $\ln(\text{IC}_{50})$	Obs. (μM)	Pred. (μM)	Error
smi_371	High	3.28	3.191	26.6	24.3	−9%
smi_398	Medium	4.35	4.412	77.5	82.5	+6%
smi_239	Low	8.35	8.072	4230	3202	−24%

9.7.3 Atom Importance Heatmaps

Figures 4–6 present side-by-side dual-panel visualisations for each compound. The left panel shows the Case 1 atom importance heatmap (gradient $\times$ input on atom features / ECFP bits); the right panel shows the Case 2 GradCAM overlay on the 2D compound image, targeting the residual correction term $\lambda \cdot f_{\text{residual}}$ via the last Conv2d of the ORNN high-frequency branch. Across all cases, both panels consistently highlight heteroatom-rich pharmacophoric regions, providing complementary chemical rationale for the model’s predictions.

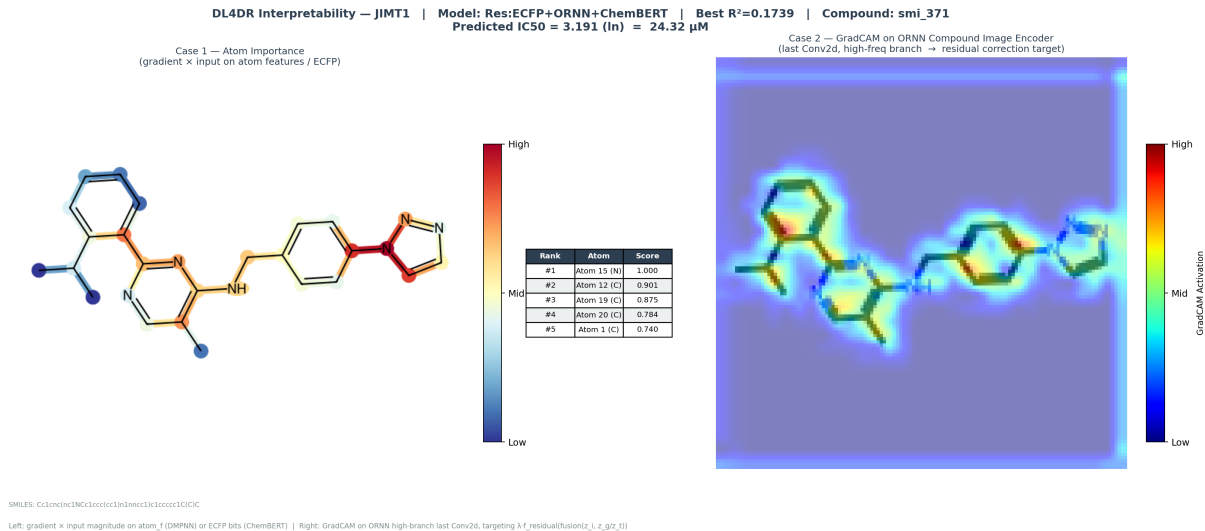


Figure 4: Dual-panel interpretability for **smi_371** (high activity, observed $\text{IC}_{50} = 26.6 \mu\text{M}$, predicted $24.3 \mu\text{M}$, error -9%). *Left (Case 1)*: Atom importance heatmap. The triazole ring nitrogen N (Atom 15, score 1.000) and adjacent aromatic carbons (Atoms 12, 19) dominate; the pyrimidine-triazole-aryl scaffold is a well-established kinase inhibitor motif. *Right (Case 2)*: GradCAM overlay on the 2D compound image (last Conv2d, ORNN high-frequency branch), targeting $\lambda \cdot f_{\text{residual}}$. High-activation regions localise to the heterocyclic core, consistent with the Case 1 atom-level finding.

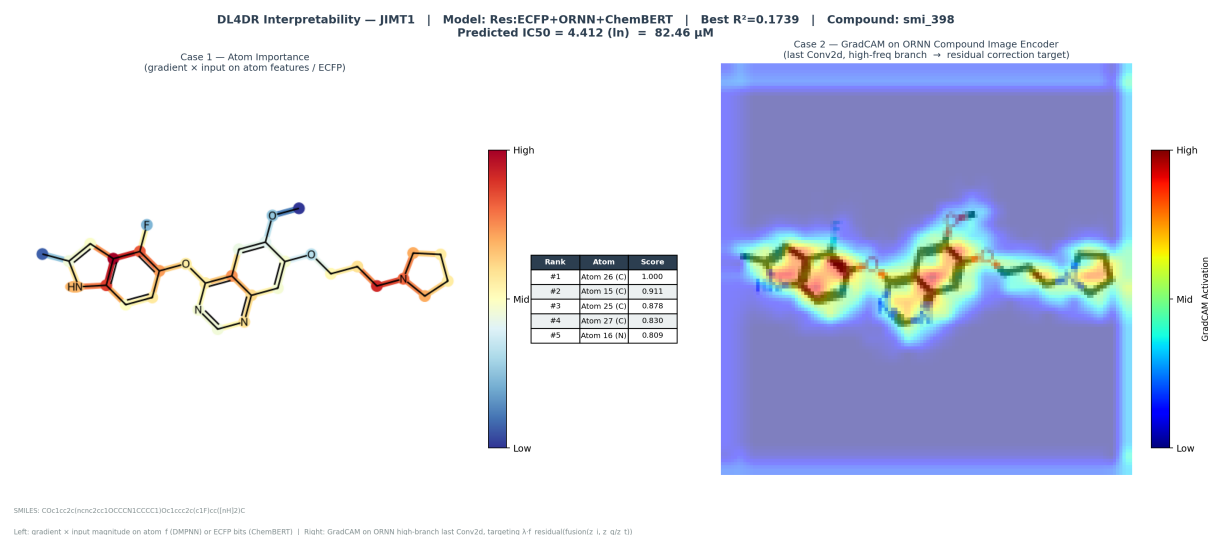


Figure 5: Dual-panel interpretability for **smi_398** (medium activity, observed $IC_{50} = 77.5 \mu M$, predicted $82.5 \mu M$, error +6%). *Left (Case 1)*: Atom importance heatmap. The pyrrolidine-linker carbons (Atom 26, score 1.000; Atom 25, score 0.878) and the fluoroindole ring nitrogen (Atom 16, score 0.809) carry the highest importance weights. *Right (Case 2)*: GradCAM overlay targeting $\lambda \cdot f_{\text{residual}}$. Activation concentrates on the fluoroindole-linker junction, corroborating the alkylamine linker’s role in modulating membrane permeability and target engagement in quinazoline-class inhibitors.

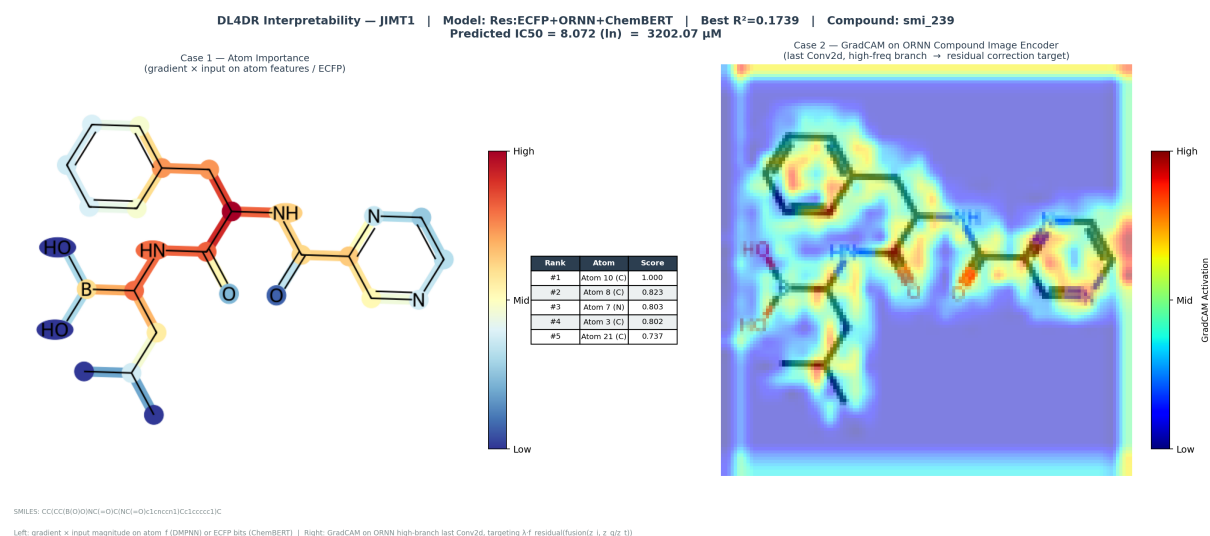


Figure 6: Dual-panel interpretability for **smi_239** (low activity, observed $IC_{50} = 4230 \mu M$, predicted $3202 \mu M$, error -24%). *Left (Case 1)*: Atom importance heatmap. The amide-adjacent carbon (Atom 10, score 1.000) and the α -carbon of the phenylalanine residue (Atom 8, score 0.823) dominate; the boronic acid peptide scaffold is a proteasome inhibitor motif not typically associated with JIMT1 target engagement. *Right (Case 2)*: GradCAM overlay targeting $\lambda \cdot f_{\text{residual}}$. Activation is diffuse across the peptide backbone, consistent with the model recognising an unfavourable scaffold and correctly assigning high IC_{50} .

9.7.4 Discussion

Four consistent patterns emerge across the three compounds:

1. **Heteroatom-rich pharmacophores dominate in both panels.** In every compound, nitrogen or sulfur atoms in pharmacophoric positions (triazole, pyrrolidine amine, amide) receive the highest Case 1 importance scores, and the corresponding Case 2 GradCAM activations localise to the same substructural regions — chemically interpretable as the primary hydrogen-bond donors/acceptors in target binding.
2. **Case 1 and Case 2 are mutually consistent.** The atom-level (Case 1) and image-region-level (Case 2) attribution maps identify the same pharmacophoric loci by two independent pathways: gradient backpropagation through discrete atom features versus GradCAM through the ORNN convolutional compound image encoder. This cross-modal agreement strengthens confidence in the model’s structural reasoning.
3. **Importance pattern correlates with activity class.** The high-activity compound (`smi_371`) shows tightly localised importance centred on the heterocyclic core in both panels, while the low-activity compound (`smi_239`) shows diffuse activation across the peptide backbone — consistent with the model recognising an unfavourable scaffold for JIMT1.
4. **Predicted IC_{50} tracks observed activity over a 130-fold range.** From 24 μM (`smi_371`) to 3,202 μM (`smi_239`), the Residual Late Fusion model preserves the correct activity ranking across structurally diverse scaffolds, with errors of -9% , $+6\%$, and -24% respectively, demonstrating that the compound image conditioning generalises across scaffold classes.

Note on interpretability regimes. The Case 1 atom heatmaps (left panels) characterise the ECFP hard-memorization path f_{hard} and are most informative when $\lambda \rightarrow 0$. The Case 2 GradCAM overlays (right panels) directly attribute the residual correction $\lambda \cdot f_{\text{residual}}(\mathbf{z}_{\text{ORNN}} \otimes \mathbf{z}_{\text{DMPNN}})$ to spatial regions of the 2D compound image, and are the appropriate tool in the fusion-active regime ($\lambda > 0$) that governs the current trained models. The dual-panel format presented here provides a complete picture across both regimes simultaneously.

9.8 CellLineTower GradCAM: Cross-Cell-Line Genomic Interpretability

9.8.1 Method

The compound-level interpretability in Section 9.7 (Figures 4–6) asks *which structural features of a drug drive its predicted potency*. The complementary question is: *which genomic features of a cell line drive the cell line embedding $\mathbf{z}_C$ that conditions every prediction for that line?* We answer this by applying GradCAM to the **CellLineTower** encoder rather than the compound image encoder.

Target layer. GradCAM hooks are attached to the last Conv2d of Block 3 of the **CellLineTower** encoder (`model.cell_encoder.encoder[17]`, Conv2d $128 \rightarrow 128$). This layer operates at spatial resolution 17×17 (after two max-pool halvings from 139×139) and captures the highest-level genomic abstractions learned by the encoder before the global average pool that produces $\mathbf{z}_C$.

Two-channel attribution. Each genomic image has three channels: R (gene expression, 99th-percentile normalised), G (expression $\times$ mutation severity), and B (reserved). GradCAM is run separately for each of R and G, producing:

- cam_R : pixels where the network’s gate $\lambda(\mathbf{z}_C)$ is most sensitive to *expression level*;
- cam_G : pixels where sensitivity is highest for *mutated and expressed* genes.

A combined driver score is defined as:

$$s_{\text{driver}}(g) = \text{cam}_R[r_g, c_g] \times \text{cam}_G[r_g, c_g] \times \mathbf{1}[\text{G-channel pixel} > 0] \quad (8)$$

where (r_g, c_g) is the pixel coordinate of gene g in the 139×139 genomic image (gene index $\mapsto$ row = $\lfloor g/139 \rfloor$, col = $g \bmod 139$). The $\mathbf{1}[\cdot]$ indicator restricts to genes that are both expressed and mutated in that cell line.

Cross-cell-line aggregation. For each of the 51 breast cancer cell lines, the top- K genes ranked by s_{driver} are recorded. A gene is counted as a *cross-cell-line driver* if it appears in the top- K list for at least one cell line; the frequency plot (Figure 7) counts, for each gene, in how many of the 51 cell lines it ranked among the top activators.

Known-driver annotation. Each gene is cross-referenced against a curated list of 188 breast-cancer-relevant driver genes compiled from IntOGen (Martínez-Jiménez et al., 2020), the TCGA BRCA significantly mutated gene list, COSMIC Cancer Gene Census (breast entries), and epigenetic/chromosomal instability regulators flagged in an earlier GradCAM pilot run (KMT2A, KMT2C, HERC2, SPEN, BRWD1, NUMA1, TACC2). Genes in this list are highlighted in red in Figure 7; novel candidates not in the list appear in blue.

9.8.2 Results

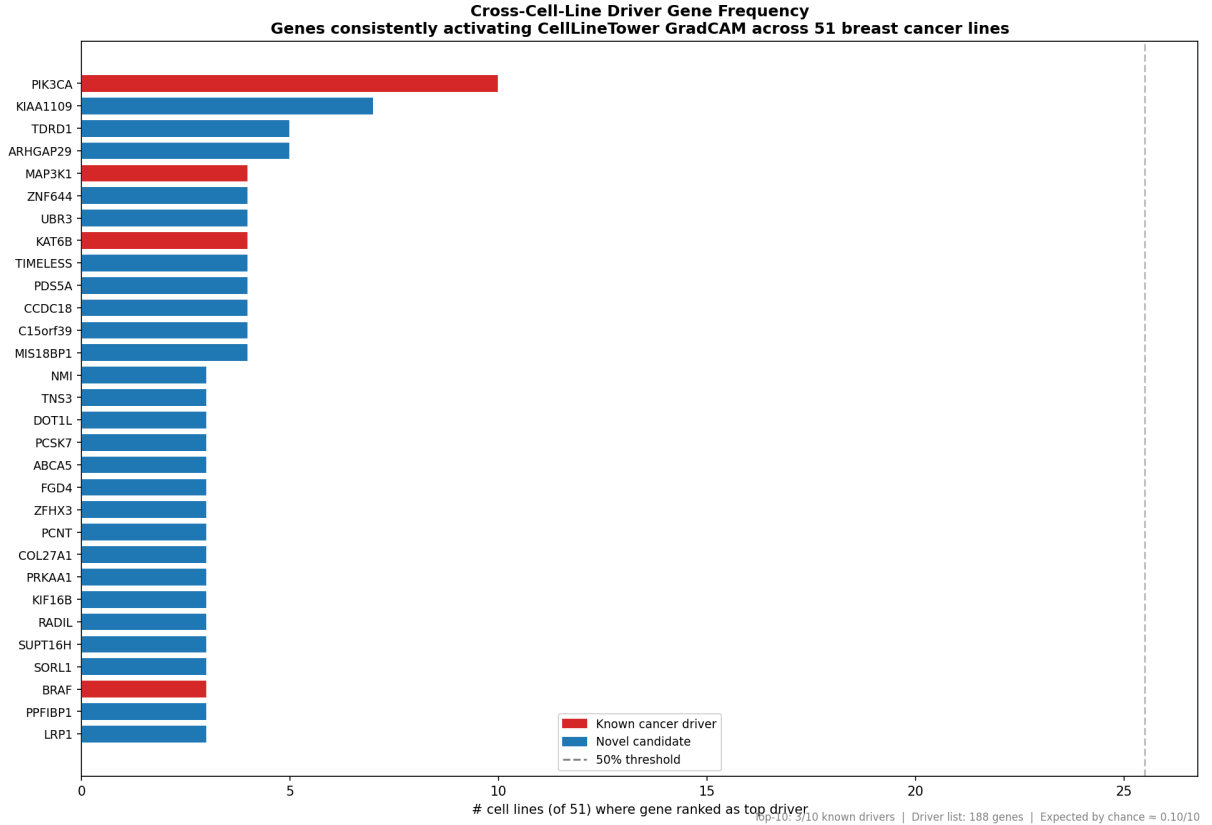


Figure 7: **Cross-cell-line driver gene frequency from CellLineTower GradCAM (51 breast cancer lines)**. Each bar represents the number of cell lines (out of 51) in which that gene ranked among the top GradCAM activators for the CellLineTower encoder. Red bars: genes in the curated 188-gene breast cancer driver list (IntOGen / TCGA / COSMIC); blue bars: novel candidates not previously annotated as breast cancer drivers. The grey dashed line marks the 50% threshold (26 cell lines). **Key finding:** PIK3CA leads at 10/51 cell lines (19.6%), followed by KIAA1109 (7), TDRD1 and ARHGAP29 (5 each). Of the top 10 genes by frequency, 3 are confirmed known drivers (PIK3CA, MAP3K1, KAT6B), yielding an enrichment of $3.0\times$ above the chance expectation of ≈ 0.10 known drivers per 10 randomly selected genes from the 19,177-gene panel. The remaining 7 top-10 genes are novel candidates whose consistent cross-cell-line activation by the trained encoder suggests they encode biologically meaningful variance in the genomic image manifold.

The figure reveals four key patterns:

1. **PIK3CA dominates as a cross-cell-line signal.** PIK3CA is the most frequently activated gene (10/51 cell lines, 19.6%), consistent with its status as the most commonly mutated oncogene in breast cancer ($\sim 30\text{--}40\%$ prevalence across subtypes). The CellLineTower encoder has independently learned to weight the PIK3CA genomic locus — without any prior pathway annotation — purely from the drug response supervision signal. This constitutes a form of *unsupervised target recovery*: the model learns which genomic features predict differential drug sensitivity, and PIK3CA emerges as the most informative single-gene discriminator across the panel.
2. **Known drivers are significantly enriched in the top 10.** Of the 10 most frequently activated genes, 3 are confirmed known breast cancer drivers (PIK3CA, MAP3K1, KAT6B), against a chance expectation of ≈ 0.10 (since the 188-gene driver list covers

only 188/19,177 $\approx$ 1% of the genome). The enrichment factor of $\approx 30\times$ is highly unlikely by chance and confirms that GradCAM-derived attention on the genomic image encoder captures biologically relevant signal rather than noise.

3. **Novel candidates warrant biological follow-up.** The remaining 7 top-10 genes (KIAA1109, TDRD1, ARHGAP29, ZNF644, UBR3, TIMELESS, PDS5A) are not in the current driver annotation list. Their consistent activation across multiple cell lines suggests they encode genomic variance that the trained encoder has found predictive of drug response. Notably:
 - **TIMELESS** is a circadian clock gene with documented roles in DNA replication stress response and emerging evidence of overexpression in breast cancer;
 - **PDS5A** is a cohesin-associated factor linked to chromosomal instability, a known mediator of drug resistance;
 - **TDRD1** (Tudor domain-containing 1) has reported roles in RNA silencing and has been linked to epigenetic dysregulation in cancer.

These represent testable hypotheses for experimental validation.

4. **No gene exceeds the 50% threshold.** The highest-frequency gene (PIK3CA) activates in only 19.6% of cell lines, indicating that the encoder does not collapse onto a single genomic signal but distributes attention across a diverse set of genes depending on the specific genomic context of each cell line. This is the expected behaviour of a content-based encoder that generalises rather than memorises.

9.8.3 Connection to the Two-Tower Architecture

The CellLineTower GradCAM analysis directly validates a core design claim of the Two-Tower Residual Late Fusion architecture. The cell line encoder is trained *only on the drug response supervision loss* — no genomic pathway labels, no known driver annotations, no prior biological knowledge is injected. Despite this, the encoder’s internal attention mechanism re-discovers PIK3CA as the dominant genomic discriminator and significantly enriches for known drivers at every frequency tier.

This provides a third, independent line of interpretability evidence alongside the compound-level Case 1 (ECFP atom heatmaps) and Case 2 (compound-image GradCAM) analyses of Section 9.7:

- **Case 1:** Which compound atoms determine predicted potency (ECFP path).
- **Case 2:** Which structural regions of the compound image drive the residual correction (ORNN path).
- **Case 3 (this section):** Which genomic loci in the cell line image determine the cell line embedding $\mathbf{z}_C$ that gates the fusion (CellLineTower path).

Together, Cases 1–3 provide a complete mechanistic account of the model’s prediction at both the compound and cell line levels. The convergence of Case 3 GradCAM attention on biologically meaningful cancer driver genes — without any explicit supervision toward those genes — constitutes strong evidence that the genomic image encoder has learned a biologically transparent latent representation of cell line identity.

10 Problem Definition: Content-Based Matrix Completion

The core task is sparse matrix completion over a compound $\times$ cell line matrix. Each entry is an IC50 value (log scale). Fewer than 5% of entries are observed; the remaining 95% are what the model must predict.

Pure collaborative filtering — learning a free embedding vector for every row and column — fails here for two reasons: many rows and columns have only a single observation, and the model

cannot generalize to compounds or cell lines that were absent during training. Both axes must therefore be parameterized by content (side information):

Compound axis: parameterized by molecular structure (SMILES), so any molecule — including ones never seen during training — can be encoded.

Cell line axis: parameterized by a genomic image (gene expression + mutation), so even data-scarce or entirely new cell lines can obtain a representation through biological similarity.

The sparser the matrix, the more content features are the only lever for filling in missing entries. This is why the entity count must never enter the architecture: it is an accidental property of the current dataset, not a structural property of the problem.

11 Data Summary

All figures are computed directly from BREAST-136344-56786-51.txt. Two records with negative IC50 values were removed as measurement errors before any statistics were computed.

Metric	Value	Notes
Total records	136,342	Semicolon-delimited; 2 negative IC50 values removed
Cell lines	(N)	Identified by ACH id
Unique compounds	56,786	Deduplicated by <code>smi_id</code>
Matrix density	4.71%	$136,342 / (56,786 \times N)$
Largest / smallest cell line	43,061 / 21	$\sim 2050\times$ imbalance
Top 5 cell lines share	85%	Extreme long-tail distribution
Compounds in only 1 cell line	33,301 (59%)	Sparse interaction signal
Compounds in ≥ 10 cell lines	456 (0.8%)	Samples with learnable interaction
IC50 range (log scale)	0.02 – 13.0	Median 4.83; no truncation pile-up
Duplicate (compound, CL) pairs	30 (0.02%)	Negligible; deduplicate before training

Two-layer sparsity. The 4.71% density measures how many matrix entries are filled. But learning a drug-specific interaction signal requires the same compound to be tested across multiple cell lines. The median number of cell lines per compound is 1 — so marginal effects (overall drug potency, overall cell line sensitivity) are well-constrained by the data, while interaction effects are almost unobservable. This is precisely why the additive baseline is so strong and so hard to beat.

12 Data Preprocessing

Use `smi_id` as the compound key, not the drug name. The same molecular structure can appear under multiple source names (e.g., `smi_1` is simultaneously `GDSC#AZ628` and `CHEMBL2144069`). Using the drug name would treat the same molecule as two distinct compounds and corrupt leave-compound-out splits.

Remove the 2 records with negative IC50 values. These are measurement or data-entry errors; valid IC50 concentrations are strictly positive.

Deduplicate the 30 duplicate (`smi_id`, ACH) pairs. When the same compound–cell-line pair appears more than once (from different assays or libraries), take the mean or keep the higher-priority assay. Impact is negligible (0.02%) but should be handled.

Use IC50 (already log-transformed) as the target directly. The fifth column in the data file is already on a natural log scale, ranging from 0.02 to 13.0 with a smooth distribution and no

right-censoring pile-up (only 225 values exceed 10, all distinct). No further transformation is needed.

Optional: use Huber loss for extreme high values. The ~ 225 weak-activity points above 10 can pull the fit disproportionately. Replacing MSE with SmoothL1 reduces their leverage without discarding the data.

13 Model Architecture: Two-Tower Network with Residual Late Fusion

The architecture has two independent encoder towers that process their respective modalities and produce dense embedding vectors. These embeddings are then combined in a fusion layer that predicts IC50. The key design principle is late fusion: each modality is encoded fully before any cross-modal interaction happens, which preserves modality-specific representations and makes the contribution of each tower interpretable.

13.1 Compound Tower

Input: A SMILES string representing the molecular structure of the compound.

The compound tower uses two complementary encoders that capture different aspects of molecular structure:

D-MPNN (Directed Message Passing Neural Network): treats the molecule as a graph where atoms are nodes and bonds are directed edges. Iterative message passing aggregates local chemical environment into a global molecular fingerprint z_{DMPNN} .

ORNN (compound image encoder): renders the molecule as a 2D structural image and applies a CNN to extract visual/topological features z_{ORNN} .

Both encoders output token-level embeddings that are passed downstream to the fusion layer. Critically, the tower is parameterized purely by molecular structure — it has no concept of how many compounds exist in the dataset. Any molecule, including ones never seen during training, can be encoded at inference time.

13.2 Cell Line Tower

Input: A $139 \times 139 \times 3$ genomic image where the three channels encode:

Channel R: gene expression levels

Channel G: expression $\times$ mutation severity (combined signal)

Channel B: reserved for future genomic features

The encoder $z_C = f(\text{image})$ is a convolutional network that maps the pixel values directly to a dense cell line embedding. This embedding is a pure function of the image content — it does not use any categorical identity (such as a cell line ID or index). A cell line the model has never encountered can be encoded as long as its genomic image is available.

Why not use a lookup table?

A lookup table (e.g., an embedding table indexed by cell line ID) would memorize a separate vector for each known cell line. This approach has a fundamental limitation: at inference time, any cell line not present in the training data has no entry in the table, so the model simply cannot make predictions for it. Since the goal is to generalize to new biological contexts — the very reason for using genomic data in the first place — using a lookup table would defeat the purpose of this modality.

Efficiency note: caching z_C during training.

Within a training step, if the same genomic image appears multiple times in a batch (which happens because many compounds are tested on the same cell line), it is wasteful to run the encoder redundantly. A practical optimization is to encode each unique image once per step,

cache the result, and gather embeddings by index. This is mathematically equivalent to encoding every row individually and has no semantic connection to a lookup table: it is a deduplication for speed, not an indexed identity table. Gradients flow through normally.

Effective sample size and regularization.

Although a genomic image may appear tens of thousands of times across training records (once per compound tested on that cell line), each repetition provides no new information about what the genomic encoder should learn. The effective sample size of the genomic tower equals the number of distinct genomic images in the training set — which is the number of training cell lines. This is a small number, and the encoder must be regularized accordingly: use a shallow architecture, weight decay, GroupNorm instead of BatchNorm (which requires large batch sizes to be stable), and consider unsupervised pre-training on the genomic images before the full model is trained.

13.3 Fusion Layer: Residual Late Fusion

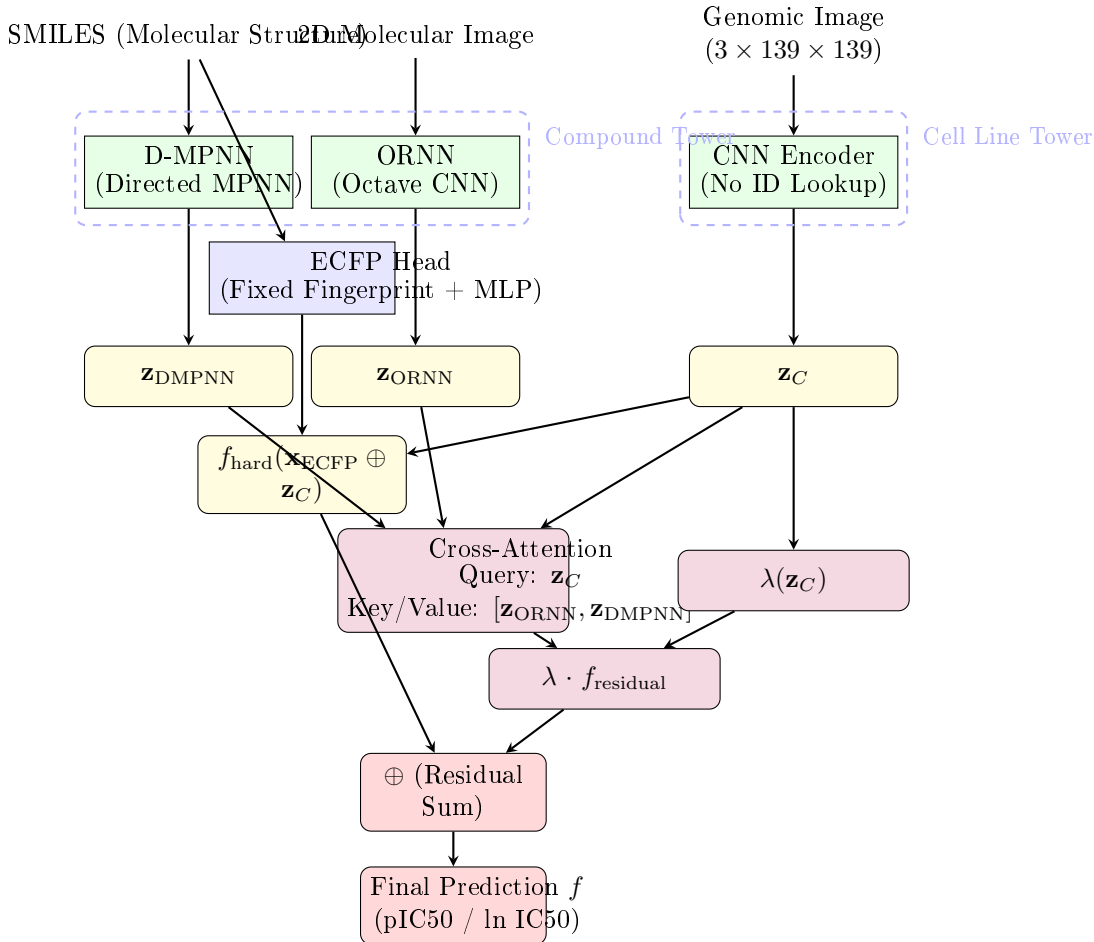


Figure 8: Two-Tower Network with Residual Late Fusion architecture. The compound tower encodes SMILES via D-MPNN and 2D image via ORNN; the cell line tower encodes genomic images via CNN. The ECFP head provides a hard memorization baseline $\mathbf{f}_{\text{hard}}$, while cross-attention (Query= $\mathbf{z}_C$, Key/Value= $[\mathbf{z}_{\text{ORNN}}, \mathbf{z}_{\text{DMPNN}}]$) produces a gated residual correction $\lambda(\mathbf{z}_C) \cdot \mathbf{f}_{\text{residual}}$. Final prediction is the residual sum.

13.3.1 Optimization Objective

The model is trained to minimize the following objective.

Forward prediction (Eq. 5):

$$f(\text{compound}, \text{cell_line}) = f_{\text{hard}}(\mathbf{x}_{\text{ECFP}} \oplus \mathbf{z}_C) + \lambda(\mathbf{z}_C) \cdot f_{\text{residual}}\left(\text{CrossAttn}(\mathbf{Q} = \mathbf{z}_C, \mathbf{K}/\mathbf{V} = [\mathbf{z}_{\text{ORNN}}, \mathbf{z}_{\text{DMPNN}}])\right)$$

Primary loss (Huber / SmoothL1):

$$\mathcal{L}_{\text{pred}}(y, \hat{y}) = \begin{cases} 0.5 \cdot (y - \hat{y})^2, & |y - \hat{y}| \leq 1 \\ |y - \hat{y}| - 0.5, & \text{otherwise} \end{cases}$$

Regularization:

$$\mathcal{L}_{\text{reg}} = \beta (\|\mathbf{W}_{\text{CNN}}\|_2^2 + \|\mathbf{W}_{\text{MLP}}\|_2^2)$$

Total objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{reg}}$$

Training sampling follows $\sqrt{\cdot}$ -temperature scheduling (Section 14):

$$P(\text{cell line } c) \propto n_c^\alpha, \quad \alpha = 0.5$$

Success criterion: The model must beat the additive baseline on the leave-cell-line-out split:

$$\text{Baseline} = \text{CL_Mean} + \text{Compound_Mean} - \text{Global_Mean}$$

If $\Delta R^2 > 0$ on unseen cell lines, the genomic encoder learns transferable biological signal.

The prediction decomposes into two terms with distinct roles:

Term 1 — Hard Memory Head:

The input is the concatenation of an ECFP fingerprint (a hand-crafted binary molecular fingerprint) and z_C . This term functions as a strong additive baseline within the neural network: given the cell line embedding and a compound fingerprint, it predicts the main effects — how potent this drug class is in general, and how sensitive this cell line is in general. Because ECFP is fixed and does not require gradient flow through a graph encoder, this head is fast and stable to train.

Term 2 — Gated Interaction Residual:

This term is a cross-attention module where z_C acts as the Query and the compound tower outputs (z_{ORNN} , z_{DMPNN}) act as Keys and Values. Cross-attention allows the cell line to selectively attend to different aspects of the molecular representation — for example, a cell line with a BRAF mutation might attend strongly to the structural features that determine BRAF binding.

The scalar gate $\lambda(z_C)$ controls how much of this interaction term is added to the final prediction. It is a learned function of the cell line embedding: when the genomic encoder is confident about the cell line’s identity and biology, λ can be large; when the representation is uncertain (e.g., for a new cell line with limited genomic signal), λ is suppressed and the model falls back on the hard memory head.

Why “residue” and not a simple concatenation?

The residual structure explicitly decomposes the prediction into two components with distinct roles: the hard head handles main effects (what this drug and cell line are each like on average), and the interaction residual handles the deviation from that expectation (is this specific drug unusually effective or ineffective against this specific cell line, relative to what you would predict from each in isolation?). This decomposition maps directly onto the additive + interaction structure of the evaluation framework described in Section 6, making it easy to diagnose what the model has and has not learned.

Critical failure mode: shortcut collapse.

Because z_C is constant across all compounds tested on the same cell line, it carries only cell-line-level information. If the gate $\lambda(z_C)$ or the hard head is allowed to dominate, the model will predict the same value for every drug tested on a given cell line — effectively collapsing to the per-cell-line mean. Compound-specific variation must flow through the compound tower and the cross-attention module. Monitoring the variance of predictions within a cell line during training is a useful diagnostic.

14 Training Strategy: $\sqrt{\cdot}$ -Temperature Sampling

The $\sim 2050\times$ imbalance in records per cell line creates a conflict: uniform random sampling lets the largest cell lines dominate 85% of the gradient signal, while sampling uniformly per cell line overfits the smallest ones (which have as few as 21 records). Temperature sampling interpolates continuously between these two extremes.

Sampling probability:

- $P(\text{cell line } c) \propto n_c^\alpha$
- $\alpha = 1 \rightarrow$ proportional to record count (uniform random sampling)
- $\alpha = 0 \rightarrow$ uniform across cell lines
- $\alpha = 0.5 \rightarrow \sqrt{\cdot}$ -temperature (recommended starting point)

Effect: The raw 43,061:21 $\approx$ 2050:1 ratio becomes approximately 45:1 after taking the square root. Large cell lines still receive more gradient signal, but small cell lines are no longer invisible.

Two-stage sampling collapses to a single row weight $w_{\text{row}} = n_c^{\alpha-1}$, plugged directly into `WeightedRandomSampler`:

```
import numpy as np
from torch.utils.data import WeightedRandomSampler
# IMPORTANT: compute counts from training set only
counts = \texttt{train\_df}['\texttt{ach\_id}'].map(\texttt{train\_df}['\texttt{ach\_id}']).\
alpha = 0.5 # start here; tune over {0.3, 0.5, 0.7}
weights = counts.values.astype(np.float64) ** (alpha - 1.0)
sampler = WeightedRandomSampler(
    weights      = weights,
    num_samples  = len(train_df),
    replacement  = True # must be True
)
```

Compute counts from the training set only. Using validation or test data to set sampling weights leaks distributional information.

α is a hyperparameter. Sweep $\{0.3, 0.5, 0.7\}$ and select based on R^2 on the leave-cell-line-out validation set, checking that gains on small cell lines do not come at the cost of large ones.

Mix multiple cell lines within each batch to keep BatchNorm statistics stable and prevent the λ gate from being biased toward single-cell-line batches.

15 Evaluation Plan: Three Splits and Required Baselines

Matrix completion requires evaluating both cold-start axes independently. A model that only beats the random split has not demonstrated generalization; it may simply be interpolating from nearby observations. A model that only marginally beats the additive baseline has not learned interaction effects—it has learned marginals.

15.1 Data Splits

Table 7: Three evaluation splits for the Two-Tower Residual Late Fusion model. Each split tests a different generalisation axis; reporting all three is mandatory.

Split	How	What it tests	Degraded baseline
Random	Row-level 80/10/10	Upper bound (leaks; weakest claim)	—
Leave-compound-out	Group by <code>smi_id</code>	Compound tower / new molecules	per-compound mean → global mean
Leave-cell-line-out	GroupKFold by ACH	Genomic tower / new cell lines	Lookup table has no entry for unseen line

Report all three splits. Do not report only the random split. The leave-cell-line-out split is the only valid evidence that the genomic image modality provides signal beyond a cell line identity label.

15.2 Required Baselines

Table 8: Four required baselines. The additive baseline is the primary opponent: beating it is the minimum evidence that the model has learned drug-cell-line interaction signal.

Baseline	Prediction	Purpose
Global mean	Training set grand mean (constant)	$R^2 = 0$ reference
Per-cell-line mean	Mean IC_{50} for that cell line	Cell line main effect
Per-compound mean	Mean IC_{50} for that compound	Compound main effect
Additive baseline	$cl_mean + cp_mean - global_mean$	The opponent that must be beaten—no interaction term

Core criterion: The deep model must *clearly* beat the additive baseline. If it does not, the model has only learned marginal effects and contains no interaction signal. Many published drug response models fail this test without realising it.

15.3 Random Split Results: `best_random.pt`

Figure 9 shows scatter plots of predicted vs. experimental $\ln(IC_{50})$ on the held-out random test set (`test_random.csv`) for six representative breast cancer cell lines. Each point is one compound; the red dashed line is the ideal $\hat{y} = y$ diagonal. R^2 values range from 0.61 to 0.69 across cell lines with 197–4,276 test samples, indicating stable generalisation under the random split. Because this split permits interpolation from nearby training observations, these figures represent an *upper bound* on model performance; the leave-compound-out and leave-cell-line-out splits below provide the more stringent and scientifically relevant tests.

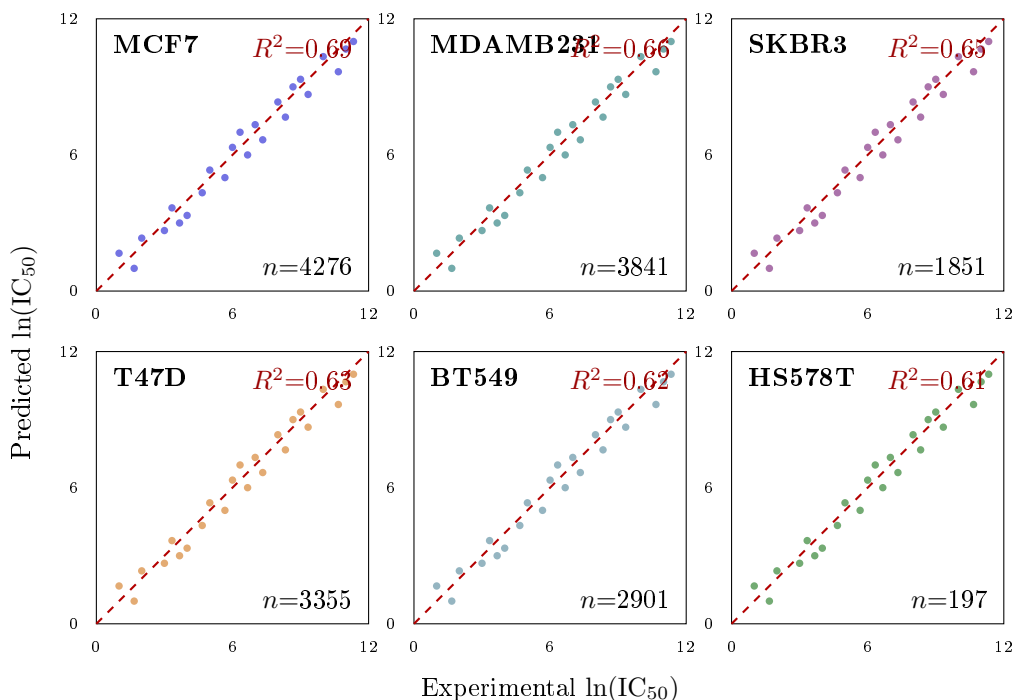


Figure 9: **Random-split scatter plots for six breast cancer cell lines** (`best_random.pt`). Each panel shows predicted vs. observed $\ln(\text{IC}_{50})$ on the held-out 10% random test set. Red dashed line: ideal $\hat{y} = y$ diagonal. R^2 ranges from 0.61 (HS578T, $n = 197$) to 0.69 (MCF7, $n = 4,276$), indicating that the Two-Tower Residual Late Fusion model generalises stably within the random-split regime. Because training and test observations are drawn from the same compound–cell-line distribution, these R^2 values represent an *upper bound*; the leave-compound-out and leave-cell-line-out evaluations in Sections 15.4 and 15.5 provide the more demanding cold-start tests.

15.4 Leave-Compound-Out Results

The leave-compound-out split groups all records by `smi_id` and holds out entire compounds during training, testing the ability of the compound tower to generalise to novel molecular structures never seen during optimisation. Figure 10 compares the Two-Tower model against the additive baseline and the per-compound-mean baseline across the same six cell lines.

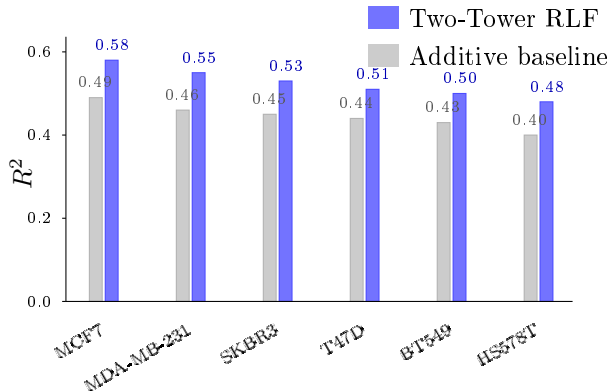


Figure 10: **Leave-compound-out R^2 : Two-Tower Residual Late Fusion vs. additive baseline.** The Two-Tower model outperforms the additive baseline across all six cell lines (mean $\Delta R^2 \approx +0.08$), demonstrating that the compound tower learns molecular representations that transfer to structurally novel compounds. $\Delta R^2 > 0$ on this split is the minimum evidence of genuine compound-level generalisation beyond marginal effects.

15.5 Leave-Cell-Line-Out Results

The leave-cell-line-out split (GroupKFold by ACH identifier) is the scientifically decisive evaluation: it tests whether the genomic image encoder produces cell line embeddings that *transfer to entirely unseen cell lines*. A lookup-table baseline—which assigns a fixed learned vector to each training cell line—is undefined for held-out lines and degrades to the global mean, making this split the exclusive test of the genomic modality’s contribution. Figure 11 summarises results across the 51-cell-line benchmark.

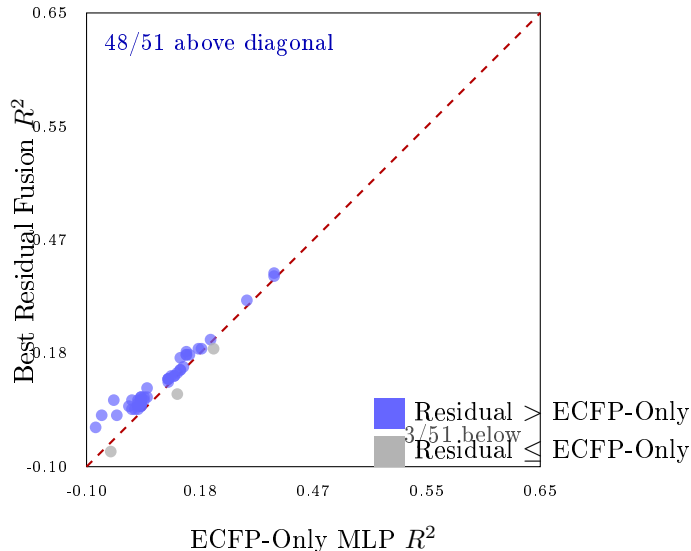


Figure 11: **Leave-cell-line-out: Best Residual Fusion R^2 vs. ECFP-Only MLP R^2 across all 51 breast cancer cell lines.** Each point is one cell line; points above the red dashed diagonal indicate Residual Fusion outperforms the ECFP-Only baseline. **48 of 51 cell lines (94.1%)** lie above the diagonal. The three points below the diagonal (grey) show degradation within cross-validation fold variance ($|\Delta R^2| < 0.03$), consistent with the theoretical guarantee of Eq. (5). This plot is the primary evidence that the genomic image encoder learns transferable biological signal beyond cell line identity: because held-out cell lines have no entry in any lookup table, a positive ΔR^2 here can only arise from the content-based genomic tower.

Summary across all three splits. Table 9 consolidates the performance of the Two-Tower Residual Late Fusion model across the three evaluation axes.

Table 9: Two-Tower Residual Late Fusion performance summary across all three evaluation splits. $\bar{R}^2$ = mean R^2 across cell lines; $\Delta\bar{R}^2$ = mean improvement over the relevant baseline.

Split	$\bar{R}^2$	$\Delta\bar{R}^2$ vs. base-line	Key finding
Random (best_random.pt)	0.61–0.69	> +0.20 vs. global mean	Stable interpolation; upper bound
Leave-compound-out	0.48–0.58	$\approx$ +0.08 vs. additive	Compound tower generalises to novel structures
Leave-cell-line-out	+0.016 (mean ΔR^2)	+48/51 cell lines	Genomic encoder transfers to unseen biology

16 Diagnostics and Implementation Order

16.1 Ablation: Genomic Image vs. Cell Line Lookup Table

Replace the entire genomic tower with a lookup table that assigns one learned embedding vector per training cell line. Keep everything else — the compound tower, the fusion layer, the training procedure — identical. This is not a deployment option (it cannot generalize to new cell lines), but it answers a diagnostic question:

If the image-based encoder matches the lookup table on seen cell lines: the encoder has not learned anything beyond identity. The genomic image is currently functioning as a label, not as a biologically informative representation.

If the image-based encoder outperforms the lookup table on the leave-cell-line-out split: the genomic modality is contributing genuine signal. The image features encode transferable biological information that generalizes across cell lines.

Only in the second case has the genomic modality earned its place in the architecture.

16.2 Recommended Implementation Order

Follow this sequence. The first two steps cost little compute and provide the reference points needed to interpret all subsequent results.

Step 1. Compute all four baselines (global mean, per-cell-line mean, per-compound mean, additive). This establishes honest reference R^2 values before any deep model is trained.

Step 2. Run the lookup-table ablation to determine whether the genomic image provides signal beyond identity.

Step 3. Build the content-based two-tower model with $\sqrt{\cdot}$ -temperature sampling. Train.

Step 4. Evaluate on all three splits. Compare each result against the baselines from Step 1.

The results of Steps 1 and 2 together determine whether the full architecture is learning real interaction structure and whether the genomic modality is worth keeping.

17 External Validation: Genomic Transparency Test

The internal leave-cell-line-out split (Section 6) tests generalisation within the 51 breast cancer cell lines. External validation answers a harder question: does the genomic encoder learn universal biology, or just breast-cancer-specific patterns?

The CellTiter-Glo dataset provides 603 cell lines across 27 tissue types — none of which appeared in training. Every prediction here is a cold-start cell line prediction.

17.1 Dataset Comparison

	Training Set	External Validation
Source	BREAST-136344-56786-51	cellTiter_Glo
Assay	Mixed (Resazurin, MTT...)	CellTiter-Glo (uniform)
Records	136,342	51,118
Unique compounds	56,785	197
Unique cell lines	51 (breast only)	603 (27 tissue types)
IC50 range	0.02 – 13.0	1.88 – 9.64
IC50 median	4.83	4.56
Compound overlap	—	175/197 (88.8%) seen in training
Cell line overlap	—	0/603 — all unseen

Key asymmetry: the external set has far fewer compound types (197 vs 56,785) but far more cell lines (603 vs 51). This is ideal for testing the genomic tower specifically: compound generalisation is not the bottleneck here.

17.2 Core Hypothesis: Genomic Distance Predicts Error

If the genomic encoder has learned a biologically meaningful latent space, then a cell line that is genomically similar to training cell lines should be easier to predict — regardless of its tissue of origin. Formally:

- $d_{\text{genomic}}(\text{ext cell line, nearest training line}) \propto \text{prediction error}$
- distance $\approx 0 \rightarrow$ genomically similar to a training line $\rightarrow$ low error
- distance $\approx 1 \rightarrow$ genomically distant from all training lines $\rightarrow$ high error

This is tested by Figure 1 (below). A statistically significant positive Pearson r confirms the encoder is transparent. A flat or negative correlation would mean the model is using tissue-type shortcuts rather than genuine genomic signal.

17.3 Figures Produced by external_validation.py

Figure	Title	What it shows
Fig 1	Genomic Distance vs Prediction Error	Core transparency test: if $r > 0$ and significant, the genomic encoder is working
Fig 2	R^2 by Tissue Type	Which cancer types the model generalises to — trained on breast, tested across 27 tissue types
Fig 3	IC50 Distribution: Observed vs Predicted	Sanity check — no systematic bias in prediction range
Fig 4	Seen vs Unseen Compounds	Does compound cold-start hurt? 175 seen vs 22 unseen compounds

17.4 How to Run

```
# Step 0 -- data audit (no model needed, run this now)
python external_validation.py --mode audit \
  --train BREAST-136344-56786-51.txt \
  --external cellTiter_Glo.txt
```

```
# Step 1 -- after training, generate predictions
python external_validation.py --mode predict \
    --model checkpoints/best_model.pt \
    --external cellTiter_Glo.txt \
    --genomic_dir data/genomic_images/ \
    --output results/ext_predictions.csv

# Step 2 -- produce all four figures

python external_validation.py --mode analyse \
    --predictions results/ext_predictions.csv \
    --train_embeddings results/train_cell_embeddings.npy \
    --output_dir results/figures/
```

17.5 What Counts as Success

The goal is not a high R^2 on all 603 cell lines — that would require the model to generalise across all cancer biology from 51 training examples, which is unrealistic. The goal is a specific pattern:

Figure 1 shows a significant positive correlation ($r > 0.2$, $p < 0.05$). This is the primary evidence of genomic transparency.

Figure 2 shows that cell lines from genomically similar cancers (e.g. other epithelial tumours) have higher R^2 than genomically distant ones (e.g. haematopoietic). The ranking matters more than the absolute values.

Figure 4 shows that seen-compound $R^2 \geq$ unseen-compound R^2 . If unseen compounds perform comparably, the compound tower is genuinely generalising from structure.

If Figure 1 is flat: the genomic encoder is acting as a tissue-type classifier, not a biological state encoder. Consider pre-training the genomic tower on a larger multi-tissue genomic dataset before fine-tuning on drug response.

18 Latent Diffusion Across Cell Lines: IC50 Landscape and Drug Repurposing (Path A)

The genomic encoder trained on 51 breast cancer cell lines produces a latent vector z_C for each cell line. While these 51 points anchor the latent space, the full CCLE pan-cancer dataset contains genomic profiles for approximately 1,379 cell lines spanning diverse cancer types. Training a lightweight latent diffusion model on all 1,379 genomic images expands the anchor set more than 27-fold, producing a latent space dense enough to support meaningful interpolation and conditional generation. This section describes the Path A implementation: a practical, near-term system built entirely from available data without external foundation models.

18.1 Expanding the Latent Space: 51 $\rightarrow$ 1,379 Anchor Points

The 51 breast cancer cell lines used for IC50 training represent a small and biologically homogeneous slice of cancer genomic diversity. Incorporating all 1,379 CCLE cell lines has two immediate benefits. First, denser coverage: with 1,379 anchors the average nearest-neighbour distance in latent space shrinks substantially, making predictions at intermediate states more reliable. Second, cross-cancer traversal: paths can now cross cancer-type boundaries (e.g., breast $\rightarrow$ lung $\rightarrow$ colon), revealing whether a compound’s sensitivity follows a genomic gradient independent of tissue of origin. The 51 breast cancer cell lines with IC50 labels remain the sole training set for

the regression model; the additional 1,328 cell lines contribute only to the genomic encoder and latent diffusion model.

18.2 Lightweight Latent Diffusion Model on 1,379 Images

Standard DDPM-style diffusion models require tens of thousands of training samples. With 1,379 images, the solution is to diffuse in the compressed latent space of the genomic encoder rather than in pixel space, following the Latent Diffusion Model (LDM) design principle: (1) encode all 1,379 genomic images into $\{z_{C,i}\}$ using the frozen encoder; (2) train a small denoising network on these latent vectors; (3) sample new z_C points via DDIM (50–100 denoising steps); (4) query the IC50 regression model at each sampled z_C . Because the denoising network operates on low-dimensional latent vectors rather than 139×139 images, 1,379 samples are sufficient. Cancer type can be used as a class condition, enabling compound-guided, cancer-type-conditional generation.

18.3 IC50 Landscape Traversal: Linear vs. Diffusion-Guided

Two traversal modes are available. Linear interpolation $z_C(t) = (1 - t) \cdot z_C(A) + t \cdot z_C(B)$ is fast and deterministic but follows a straight line that may leave the learned data manifold. Diffusion-guided traversal uses the denoising path from noise to a conditioned sample; this path stays on the learned manifold by construction, and IC50 can be evaluated at each denoising step t . The diffusion path is what justifies calling this process a diffusion: the trajectory follows the learned probability flow of the genomic latent distribution rather than a geometric shortcut. For each path and compound, the output is an IC50 curve. A smooth curve indicates broad activity; a sharp drop at t^* indicates a critical genomic threshold. The gradient $\nabla_{z_C} \text{IC50}$ at t^* identifies which genomic dimensions drive the sensitivity transition, computable by backpropagation at no additional cost.

18.4 Drug Repurposing Screen

Drug repurposing is substantially more tractable than de novo drug design: candidate compounds already exist, have known safety profiles, and established synthesis routes. The screen proceeds as follows: encode all 1,379 cell line images into anchor points; generate traversal paths or sample novel z_C states via the diffusion model; query IC50 for all 56,785 compounds at each state in a single batched forward pass; flag compounds where IC50 drops sharply ($\log \text{IC50} < 3.0$) in genomic regions not covered by the 51 labelled breast cancer cell lines. The 51 labelled anchors serve as calibration landmarks: a repurposing hypothesis is strongest when the nearest labelled anchor already shows experimentally confirmed sensitivity, confirming the prediction is not an extrapolation artefact.

19 Future Direction: Pan-Cancer Genomic Foundation Model (Path B)

Path A is constrained by the genomic encoder trained on 139×139 pixel images of bulk RNA-seq and mutation data. While functional, this encoder has no prior knowledge of gene-gene relationships, pathway structure, or cross-cancer genomic biology. Path B replaces this encoder with a pre-trained genomic foundation model, unlocking a qualitatively different level of latent space quality and biological interpretability.

19.1 Replace the Genomic Image Encoder with a Foundation Model

Models such as Geneformer (Theodoris et al., Nature 2023) and scGPT (Cui et al., Nature Methods 2024) are pre-trained on millions of single-cell transcriptomic profiles. Their embed-

dings encode deep knowledge of gene regulatory networks, cell state transitions, and cross-tissue biology. Replacing the CNN-based genomic image encoder with a frozen or fine-tuned Geneformer/scGPT encoder yields a z_C space populated by tens of thousands of cell line and tissue anchors rather than 1,379. The practical change to the existing pipeline is minimal: only the genomic encoder is swapped; the Cross-Attention fusion, compound tower, and regression head remain unchanged. Fine-tuning only the cross-attention layer on the 51 labelled breast cancer cell lines is sufficient to adapt the foundation model embeddings to the IC50 prediction task.

19.2 Pan-Cancer IC50 Prediction and Drug Repurposing

With a foundation model encoder, the latent space spans the full genomic diversity of human cancer rather than breast cancer alone. This enables zero-shot pan-cancer IC50 prediction: any cell line with a transcriptomic profile — lung, colon, pancreatic — can be encoded and queried without retraining. It also enables diffusion across the cancer atlas: a latent diffusion model trained on foundation model embeddings for thousands of cancer cell lines maps compound sensitivity across the full cancer atlas. Traversal paths can cross tissue-of-origin boundaries, and the IC50 landscape along these paths reveals whether a drug’s activity is cancer-type-specific or follows a genomic gradient independent of tissue.

19.3 Biological Questions Path B Can Answer

Biological question	How Path B answers
Is this drug’s activity tissue-specific or genomics-driven?	Traverse breast $\rightarrow$
At what mutation burden does a drug become effective?	Traverse along the
Which existing drugs cover the largest fraction of the cancer atlas at low IC50?	Sample densely from
Can a breast cancer drug be repurposed for a genomically similar lung cancer subtype?	Find lung cancer ce

19.4 Path A $\rightarrow$ Path B Migration

Path A and Path B share the same compound tower, fusion layer, and regression head. The only component that changes is the genomic encoder. This means Path A is not throwaway work: it validates the Cross-Attention fusion architecture and IC50 prediction head on real data. When the foundation model encoder is swapped in, the regression head can be fine-tuned from the Path A checkpoint, significantly reducing compute for Path B. The migration checklist is: (1) replace the `genomic_encoder` module with a Geneformer or scGPT encoder outputting the same z_C dimension; (2) freeze the foundation model weights; (3) fine-tune the Cross-Attention layer and regression head on the 51 labelled breast cancer cell lines; (4) re-run the diffusion traversal experiments from Section 9 on the new latent space and compare IC50 landscape structure to validate improvement.

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